DataForge

Project Report submitted

in partial fulfilment of the requirements for the degree

of

Bachelor of Technology in

Computer Science & Engineering

by

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CERTIFICATE

This is to certify that the project report entitled “DataForge” has been completed and submitted by Mohammad Numan Narwaroo *(CUS-22-SET-10320)*, Mohammad Usman Dar *(CUS-22-SET-10314)*, Faheem Ahmad Bhat *(CUS-22-SET-10331)* , Mubashir Shabir *(CUS-22-SET-10309)* to the Department of Computer Science and Engineering, Government College of Engineering and Technology (GCET), Safapora, Ganderbal, J&K, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology (B.Tech.) in Computer Science and Engineering.

This project is a bonafide record of work carried out by the above-mentioned students under my guidance and supervision during the academic session **2025–2026**. The work presented in this report is original and has not been submitted, either in whole or in part, to any other University or Institution for the award of any degree, diploma, or certificate.
I am satisfied with the quality of work, dedication, and effort demonstrated by the students during the development of this project and recommend this report for evaluation.

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STUDENT DECLARATION

We declare that this project report titled “DataForge” submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering is a record of original work carried out by us under the supervision of Dr. Parvaz Ahmad Malla.

We further declare that this project report is a result of our own work carried out under the guidance of our supervisor and represents our sincere efforts towards the successful completion of the project.

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ABSTRACT

Data has become a critical asset for organisations, researchers, and businesses, supporting decision-making. However, extracting meaningful insights from raw datasets often requires expertise in data analytics, programming, and machine learning, creating a significant barrier for users who need data-driven insights but lack technical knowledge.

To address this challenge, there is a growing need for intelligent systems that can simplify the analytical workflow and make advanced data analysis accessible to a broader audience. A key part of this process is proper data preparation, since raw datasets often contain missing values, inconsistencies, duplicates, and formatting issues that can affect the quality of analysis. Therefore, any modern analytics platform must not only analyse data but also ensure that it is properly prepared for further processing.

DataForge is a web-based automated analytics platform designed to simplify the entire data analytics lifecycle. The platform incorporates a streamlined data preparation pipeline that enables datasets to be cleaned, transformed, and made analysis-ready with minimal user intervention. It supports the complete analytical workflow, beginning with data ingestion and preparation, followed by automated data cleaning, exploratory data analysis, intelligent insight generation, natural-language interaction, and predictive modelling, all within a unified environment.

By leveraging modern technologies and artificial intelligence, DataForge streamlines the analytics process through interactive visualisations, automated insights, and machine learning capabilities. Users can explore their data, ask queries in natural language, and build predictive models with minimal manual effort. This significantly reduces the complexity associated with traditional analytics tools while improving accessibility, efficiency, and usability.

The platform is intended for small to medium-sized organisations, students, researchers, analysts, and those seeking to make data-driven decisions more effective. By combining automation, machine learning, and a user-friendly design, DataForge transforms raw data into actionable knowledge and meaningful business insights.

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ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
AutoML	Automated Machine Learning
CSV	Comma-Separated Values
EDA	Exploratory Data Analysis
HTML	HyperText Markup Language
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LLM	Large Language Model
ORM	Object Relational Mapping
PWA	Progressive Web Application
REST	Representational State Transfer
SQL	Structured Query Language
UI	User Interface
URL	Uniform Resource Locator

CHAPTER 1
INTRODUCTION

1.1 Data

In the digital era, data has become one of the most valuable resources for organisations, businesses, researchers, and institutions. Every day, enormous amounts of structured and unstructured data are generated through online transactions, customer interactions, social media platforms, financial systems, healthcare records, and operational activities. This rapid growth of data has created significant opportunities for organisations to improve decision-making, identify trends, predict future outcomes, and optimise business operations.

Data analytics plays a crucial role in extracting meaningful insights from raw datasets. Modern organisations rely heavily on analytical tools and machine learning techniques to understand customer behaviour, monitor performance, detect anomalies, and support strategic planning. However, the process of data analysis often involves multiple complex stages such as data preprocessing, visualisation, statistical analysis, and predictive modelling.

Although advanced analytical platforms and business intelligence tools are available, many of them require technical expertise in programming languages, database systems, and machine learning frameworks. Small businesses, students, researchers, and non-technical users frequently face difficulties in utilising these tools effectively due to their complexity, high cost, and steep learning curve.

With the advancement of artificial intelligence and automated machine learning technologies, there is an increasing need for intelligent platforms that can simplify the analytical workflow and make data-driven decision-making more accessible to a broader audience.

Consequently, there is a growing demand for intelligent and user-friendly analytics platforms

that can automate complex analytical processes and make data-driven decision-making accessible to a wider audience.

1.2 Background

The increasing availability of data has led to the widespread adoption of analytics and business intelligence tools across various industries. Organisations use these tools to extract insights, monitor performance, identify trends, and support strategic decision-making. As a result, several platforms and technologies have been developed to assist users in processing, analysing, and visualising large datasets.

Despite these advancements, transforming raw data into meaningful information remains a challenging task. Data analysis workflows typically involve multiple stages such as data preprocessing, visualisation, statistical interpretation, and predictive modelling. These processes often require expertise in programming, analytical frameworks, and machine learning techniques.

Consequently, many students, researchers, small businesses, and non-technical users face difficulties in effectively utilising their data. The lack of accessible and integrated analytical systems increases dependency on technical specialists and limits the broader adoption of data-driven decision-making. These challenges highlight the need for an intelligent analytics platform that simplifies the complete analytical workflow while maintaining the effectiveness of modern data analysis techniques.

1.3 Problem Statement

Despite the availability of modern analytical tools, there remains a significant gap between raw data availability and accessible data-driven decision-making for non-technical users. Existing platforms either require advanced programming knowledge or lack integrated automation for data cleaning, insight generation, machine learning, and natural language interaction.

As a result, many users struggle to transform raw data into meaningful insights efficiently. This creates delays in decision-making, increases dependency on technical specialists, and limits the adoption of advanced analytics in small and medium-scale environments.

1.4 Objectives

The objectives of the system are as follows:

(i) To develop a web-based analytics platform for efficient dataset management, processing, and analysis.

(ii) To automate preprocessing and exploratory data analysis (EDA) for generating meaningful insights and visualisations.

(iii) To provide AI-assisted natural language interaction and automated machine learning capabilities for data exploration and predictive analytics.

(iv) To generate analytical reports that support effective data-driven decision-making.

1.5 Scope

DataForge is intended for small to medium-sized organisations, students, researchers, analysts, and those who require accessible and efficient data analysis. The system enables users to upload CSV datasets, perform automated data cleaning, generate visualisations and insights, interact with data through natural language queries, and build predictive models using AutoML techniques.

The project aims to simplify the data analytics workflow and reduce dependency on programming expertise, making data-driven decision-making more accessible to a wider range of users.

1.6 Target Users

DataForge is designed to serve a diverse range of users who require data analysis and decision-support capabilities without extensive expertise in programming, machine learning, or data science. The platform aims to simplify the analytical workflow by providing automated data processing, visualisation, insight generation, and predictive modelling features. The target users of the system are presented in Table 1.1.

Table 1.1: Target User Segments and their primary use cases

User Segment	Primary Use Case
Business Analysts	Sales, revenue, and KPI analysis without Python dependency
Academic Researchers	Exploratory analysis and model prototyping for thesis work
Product Managers	User behaviour and funnel analysis from exported CSVs
Students	Learning data science concepts through automated analytics and machine learning workflows.
SME Decision Makers	Reporting on business metrics

Table 1.1 illustrates the user groups for which DataForge is designed and the key applications of the platform for each category. The system caters to both technical and non-technical users by providing automated analytics, visualisation, insight generation, and machine learning functionalities through an user-friendly interface. This enables users to derive meaningful information from data and make informed decisions with minimal manual effort.

CHAPTER 2
LITERATURE REVIEW

The rapid growth of data across industries has increased the need for efficient tools capable of transforming raw data into meaningful insights. To address the challenges associated with data analysis, several platforms and technologies have been developed that simplify data preparation, visualisation, and decision-making processes.

Google Colab is a cloud-based Python development environment developed by Google that enables users to perform data analysis, machine learning, and deep learning tasks through Jupyter notebooks. It provides free access to computational resources and has become widely used in academic and research environments for developing and testing data-driven applications.

Microsoft Power BI is a business intelligence and data visualisation platform that allows users to create interactive dashboards, reports, and visual analytics. It supports connectivity with multiple data sources and helps organisations transform raw data into meaningful business insights through graphical representations and real-time monitoring.

Tableau is another widely adopted business intelligence and visualisation tool known for its powerful drag-and-drop interface. It enables users to create interactive charts, dashboards, and reports without extensive programming knowledge. Tableau is commonly used for data exploration and visualisation.

Orange3 is an open-source data mining and machine learning platform that provides a visual programming interface for data analysis. Users can perform data preprocessing, visualisation, classification, clustering, and predictive modelling through workflow-based components without writing complex code.

RapidMiner is a data science platform that supports data preparation, machine learning, predictive analytics, and model deployment. Its low-code environment enables analysts and

business users to develop analytical workflows using graphical process design, making advanced analytics more accessible.

Julius AI represents a newer generation of AI-powered analytics tools that combine data analysis with natural language interaction. Users can upload datasets and ask queries in plain English, allowing the system to generate insights, visualisations, and explanations automatically. This conversational approach significantly reduces the technical barriers associated with traditional analytics systems.

The evolution of these platforms reflects the growing demand for simplified and intelligent analytics solutions. While traditional business intelligence tools primarily focused on data visualisation and reporting, newer platforms have gradually incorporated machine learning, automation, and artificial intelligence to reduce manual effort and improve analytical capabilities. This shift has led to significant research in areas such as Automated Machine Learning (AutoML), intelligent data preparation, and conversational analytics.

In addition to these platforms, significant research has been conducted in the field of **Automated Machine Learning (AutoML)**. Thornton et al. introduced Auto-WEKA, which automated algorithm selection and hyperparameter optimisation, reducing the manual effort required in machine learning workflows. Feurer et al. further enhanced this concept by developing robust AutoML techniques capable of generating high-performing predictive models across diverse datasets. These advancements demonstrated that machine learning could be made accessible to users without extensive expertise in model development.

Data preprocessing and cleaning remain critical stages of the analytics lifecycle. Studies indicate that a significant portion of analytical effort is spent handling missing values, duplicate records, inconsistent formats, and outliers. Modern profiling tools such as YData Profiling automate dataset inspection by generating statistical summaries and identifying data quality issues, thereby improving the efficiency and reliability of data analysis.

Exploratory Data Analysis (EDA) has also evolved through the use of libraries such as Pandas and visualisation frameworks such as Chart.js. These technologies enable analysts to discover trends, correlations, and patterns within datasets through interactive charts and statistical

summaries. Effective visualisation techniques support informed decision-making and improve data interpretation.

Recent advancements in Artificial Intelligence and Large Language Models (LLMs) have introduced conversational analytics systems that allow users to interact with datasets using natural language. Such systems can generate summaries, recommendations, and analytical insights automatically, making data analysis more accessible to non-technical users.

The evolution of these platforms reflects the growing demand for simplified and intelligent analytics solutions. While traditional business intelligence tools primarily focused on data visualisation and reporting, newer platforms have gradually incorporated machine learning, automation, and artificial intelligence to reduce manual effort and improve analytical capabilities. This shift has led to significant research in areas such as Automated Machine Learning (AutoML), intelligent data preparation, and conversational analytics.

Overall, the existing literature demonstrates significant progress in business intelligence, data visualisation, automated machine learning, intelligent data preparation, and conversational analytics. However, most current solutions focus on individual stages of the analytics lifecycle rather than providing a complete end-to-end analytical environment. As a result, users often rely on multiple tools to perform data cleaning, exploration, machine learning, reporting, and insight generation. This limitation highlights the need for an integrated platform that combines these capabilities within a unified and user-friendly system.

2.1 Project Gap

Despite the advancements discussed in the previous section, several limitations still exist in current analytics platforms. Existing analytics platforms such as Tableau, Power BI, and Google Looker Studio provide strong visualisation capabilities but often require technical expertise, expensive subscriptions.

Most existing systems lack integrated support for automated data cleaning, AI-assisted querying, intelligent insight generation, AutoML-based prediction, and report generation

within a single platform. As a result, users often encounter fragmented workflows, increased complexity, and a greater dependency on technical expertise.

To address this gap, the project "**DataForge**" aims to develop a unified web-based analytics platform that combines automated preprocessing, exploratory data analysis, AI-powered insights, natural-language querying, AutoML model training, and report generation into a single user-friendly system. The platform is designed to simplify data analytics and make decision-making more accessible for small businesses, students, researchers and analysts.

2.2 Importance and Novelty

To overcome the limitations identified in the existing literature and current analytics platforms, DataForge introduces an integrated approach that combines automation, machine learning, and artificial intelligence within a single web-based environment.

The importance of DataForge lies in its ability to reduce the complexity of modern data analytics by combining multiple analytical capabilities into a single user-friendly platform. Instead of relying on separate tools for preprocessing, visualisation, machine learning, and reporting, users can perform the entire analytics workflow within one integrated system.

The novelty of DataForge is its combination of automated data preparation, exploratory data analysis, AI-assisted natural language interaction, intelligent insight generation, and AutoML-based predictive modelling in a unified web application. While platforms such as Tableau and Power BI primarily focus on visualisation, and tools like Orange3, RapidMiner, and Google Colab concentrate on specific analytical tasks, DataForge brings these capabilities together through an accessible interface designed for both technical and non-technical users.

By integrating automation, machine learning, and artificial intelligence into a single workflow, DataForge aims to improve efficiency, reduce manual effort, and enable users to transform raw data into meaningful insights with minimal technical expertise.

CHAPTER 3

SYSTEM DESIGN AND METHODOLOGY

The DataForge begins when a user uploads a dataset in supported formats, such as CSV, XLS, and XLSX files or imports data directly through Google Sheets links. The uploaded dataset is validated and processed to ensure data consistency and compatibility with the system.

After successful ingestion, the system performs automated preprocessing operations. These include handling missing values, removing duplicate records, standardising column names, and resolving data-type inconsistencies. The cleaned dataset is then prepared for further analytical processing.

The next stage involves Exploratory Data Analysis (EDA), where the system generates statistical summaries, visualisations, and descriptive insights that help users understand the characteristics and patterns present in the dataset. These visual outputs provide a quick overview of trends, distributions, and relationships among variables.

Once the data has been analysed, users can interact with it through the AI Query Module by asking queries in natural language. The system interprets the query, performs the required analysis, and returns meaningful answers along with relevant visualisations when applicable. Alternatively, users can utilise the AutoML module to build predictive models without requiring extensive machine learning knowledge.

In addition, the Insight Generation Engine identifies important trends, correlations, anomalies, and business-relevant observations from the dataset. These insights help users gain a deeper understanding of the underlying data and support decision-making.

Finally, the generated insights, visualisations, and predictive results are compiled into analytical reports that can be viewed or exported by the user. The system ensures that raw data is transformed into actionable information through an efficient, automated, and user-friendly analytical process.

To support this efficiently, DataForge is designed using a layered multi-tier architecture that separates user interaction, application logic, analytical processing, and data management functionalities.

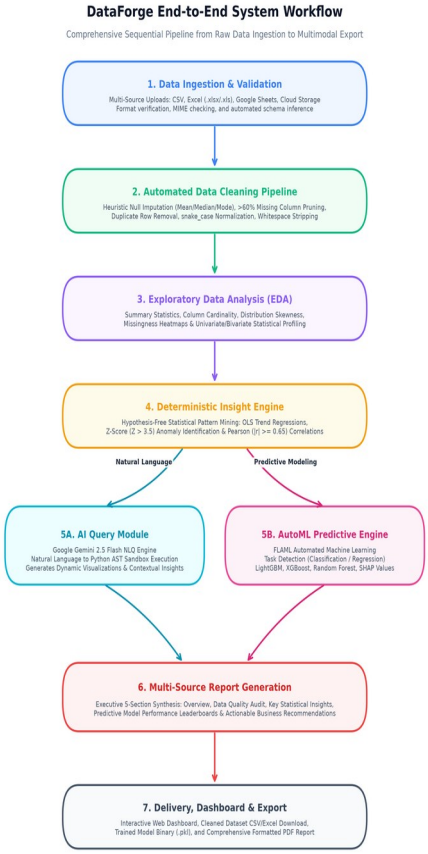


Figure 3.1: DataForge System Workflow Pipeline

Figure 3.1: System Workflow

3.1 System Architecture

DataForge follows a **layered multi-tier web architecture** that separates presentation, application, processing, and data management functionalities as shown in Figure 3.2. The **Presentation Layer** consists of HTML5 and Jinja2-based web interfaces enhanced with Bootstrap, JavaScript, Chart.js, and Socket.IO to provide an interactive and responsive user experience.

The **Application Layer** is built using the Flask framework and exposes RESTful APIs and WebSocket endpoints for handling dataset management, data cleaning, exploratory data analysis (EDA), AI-powered querying, AutoML model training, report generation, and

project management functionalities.

The **Processing and Analytics Layer** contains the core analytical components of the system, including the Data Cleaning Engine, EDA, AutoML Engine, and the Google Gemini AI integration for natural language data analysis. Long-running analytical operations are executed asynchronously using Celery workers with Redis acting as the message broker and caching layer.

The **Data and Storage Layer** comprises Supabase PostgreSQL for metadata management, Supabase Object Storage for datasets, reports, and trained models, local file storage for temporary analytical artefacts, and Redis for caching and task management. This architecture ensures scalability, modularity, efficient resource utilisation, and real-time communication between system components.

The architectural components work together to provide a modular and scalable analytical platform. The Presentation Layer facilitates user interaction, the Application Layer manages system operations and requests, the Processing & Analytics Layer performs data analysis and machine learning tasks, while the Data & Storage Layer ensures reliable storage and retrieval of datasets, reports, and analytical artefacts.

While the system architecture describes the structural organisation of the platform, it is equally important to understand how information moves between different components during execution. This is represented through the Data Flow Diagram (DFD).

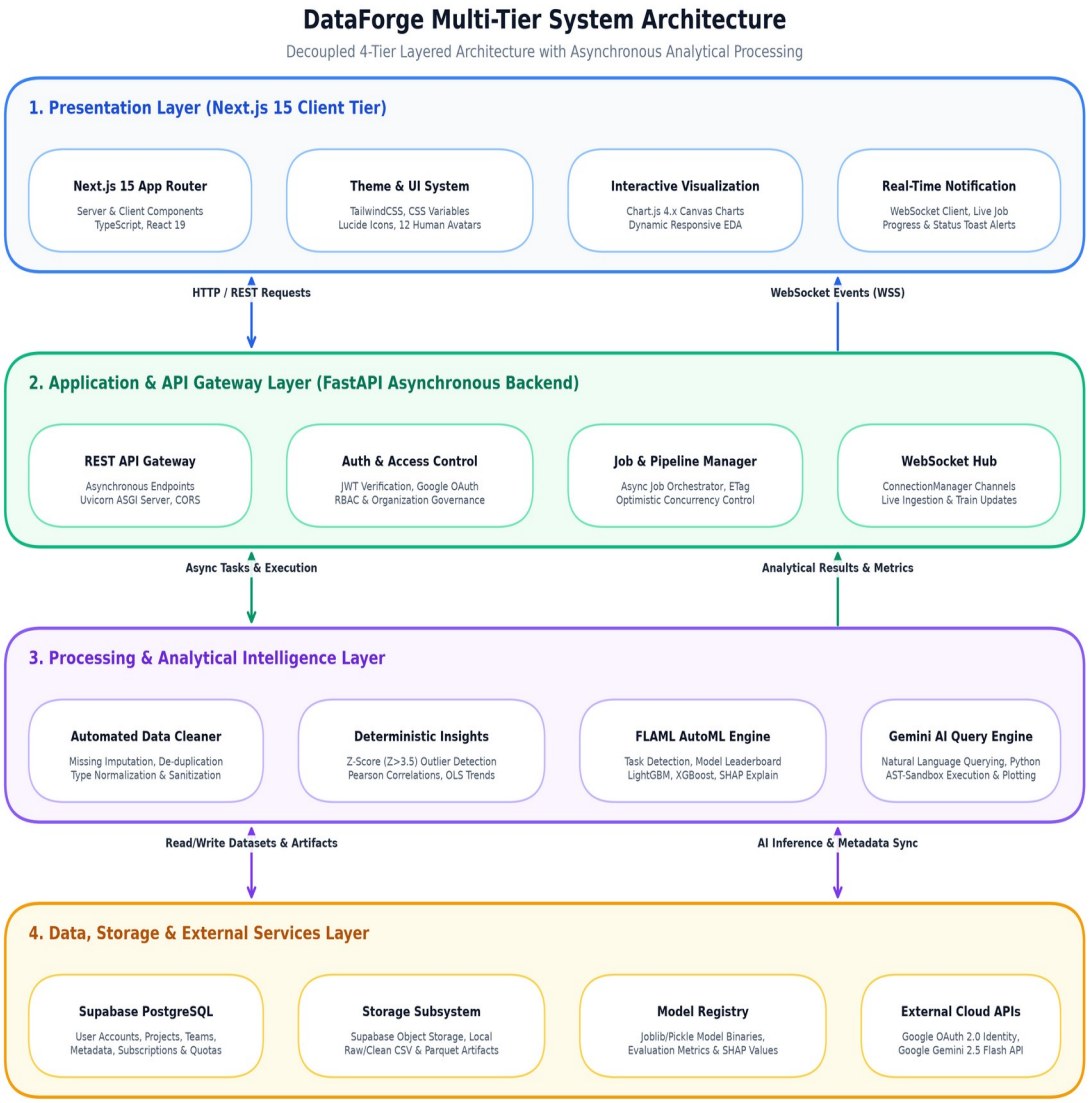


Figure 3.2: DataForge Enterprise Multi-Tier System Architecture

Figure 3.2: DataForge System Architecture

3.2 Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation of how data moves through a system. It illustrates the flow of information between users, system processes, external entities, and data stores. The primary purpose of a DFD is to provide a clear understanding of how data is collected, processed, stored, and transformed into useful outputs.

In the project, data originates from multiple sources such as CSV files, Excel files (.xls/.xlsx), and Google Sheets links provided by different users. The system processes this data through several analytical stages, including data cleaning, exploratory data analysis (EDA), AI-powered querying, automated machine learning (AutoML), insight generation, and report creation. Throughout this process, the generated information is stored and retrieved from databases and storage repositories whenever required.

The DFD helps visualise the interaction between different modules of the system and demonstrates how information flows from the user to the analytical components and finally returns as meaningful insights, visualisations, predictive models, and reports. To provide both a high-level overview and a detailed representation of the system's functionality, the Data Flow Diagram is presented in two forms: the **Level 0 DFD (Context Diagram)**, which illustrates the overall interaction between the system and external entities, and the **Level 1 DFD**, which provides a more detailed view of the internal processes and data flows within the DataForge platform.

3.2.1 Context Diagram (level 0) DFD

The Level 0 DFD represents DataForge as a single process interacting with external entities. The User uploads datasets, submits natural language queries, requests reports, and receives analytical outputs. Google OAuth provides authentication services, while Google Sheets

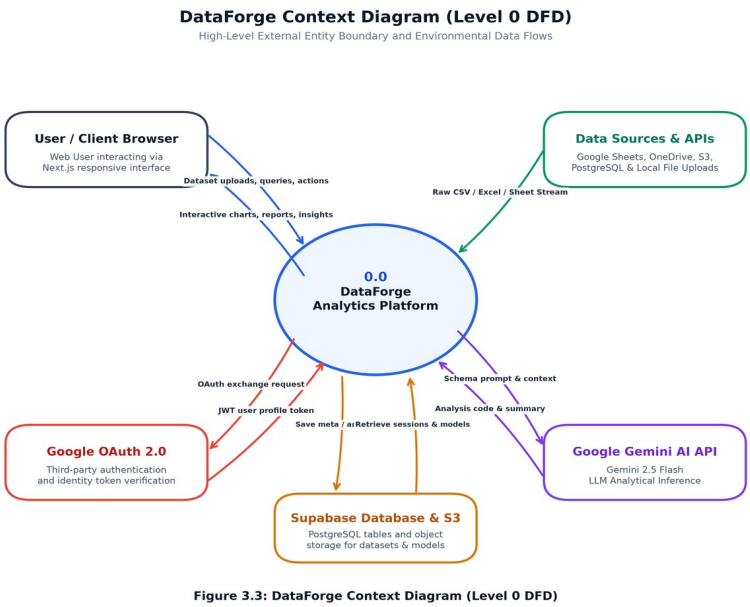


Figure 3.3: Context Diagram (level 0) DFD

serves as an external data source. The Gemini AI API supports AI-powered data analysis and query processing. Supabase stores datasets, metadata, reports, and trained models.

3.2.2 Detailed Diagram (level 1) DFD

The Level 1 DFD provides a detailed view of the internal functioning of the DataForge system. It illustrates how data flows through various analytical modules, beginning with data ingestion and cleaning, followed by insight generation, AI-powered querying, automated machine learning, and report generation. The diagram also shows the interaction of external entities such as the User/Client Browser, Google Sheets API, and Gemini API with the system. Throughout the analytical workflow, datasets, generated insights, reports, and machine learning models are stored and retrieved from the Database and Object Storage repository whenever required. This level of the DFD provides a clearer understanding of the major processing components and the movement of information within the platform.:

- P1 – Data Ingestion: Accepts datasets from CSV, Excel, and Google Sheets sources. The uploaded data is validated and prepared for further processing.
- P2 – Data Cleaning: Performs preprocessing operations such as handling missing values, removing duplicate records, standardising column names, and correcting data types to improve data quality.
- P3 – Insight Generation: Analyses the cleaned dataset using statistical and rule-based techniques to identify trends, anomalies, correlations, and other meaningful patterns.
- P4 – AI Query Module: Processes natural language queries submitted by users and utilises the Gemini AI engine to generate contextual explanations, summaries, and analytical responses based on the dataset.
- P5 – AutoML Module: Automatically trains and evaluates multiple machine learning models, selects the best-performing model, and generates predictive analytics results.
- P6 – Report Generation: Compiles insights, visualisations, AI-generated findings, and machine learning results into structured analytical reports for users.
- Database & Object Storage: Stores uploaded datasets, cleaned data, generated insights, machine learning models, reports, and other analytical artefacts required by the system.

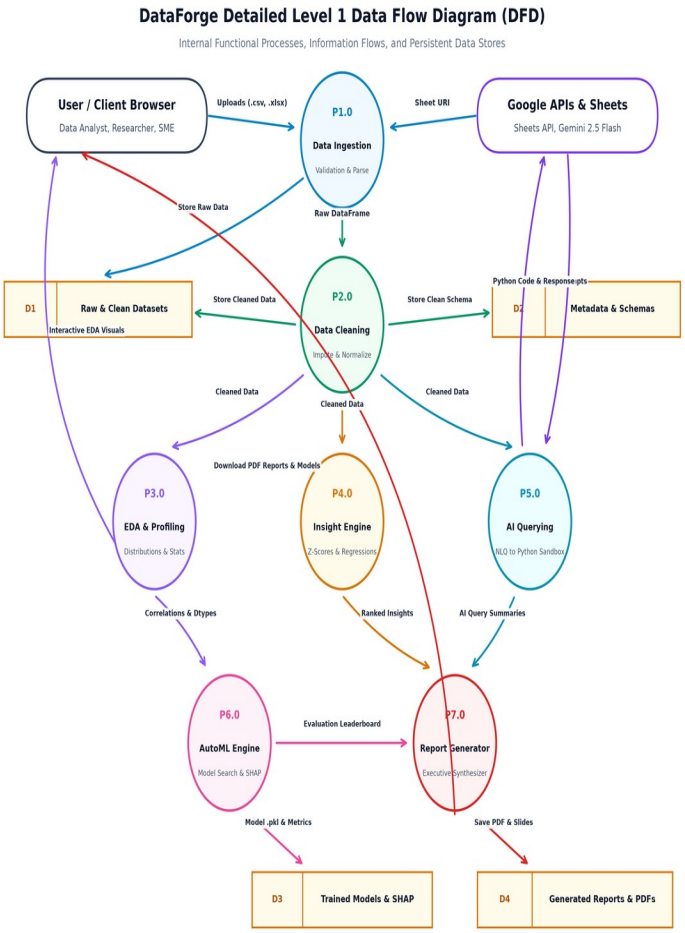


Figure 3.4: DataForge Detailed Level 1 Data Flow Diagram (DFD)

Figure 3.4: Detailed Diagram (level 1) DFD

The DFD illustrates the movement of data between different processes within the system. To understand the functionality of these processes in greater detail, the major modules of DataForge are described in the following section.

3.3 System Modules

Based on the architecture discussed earlier, the DataForge platform is organised into several

functional modules. Each module performs a specific task within the analytics pipeline and collectively contributes to transforming raw data into meaningful insights and reports. The framerwork showing different interconnected workable modules of our project is shown in Figure 3.5

3.3.1 User Authentication Module

The User Authentication Module provides secure access to the system through Google OAuth authentication. It verifies user identity, manages login sessions, and ensures that only authorised users can access project data and analytical features. This module also maintains user-specific workspaces and project information. The main functions of this module are:

i. User login and logout
ii. Session management
iii. Access control
iv. User profile management

3.3.2 Data Ingestion Module

The Data Ingestion Module is responsible for collecting datasets from various sources. It supports CSV, XLS, XLSX files, and public Google Sheets links. Uploaded datasets are validated and converted into a structured format suitable for analysis. The main functions of this module are:

i. Dataset upload
ii. File validation
iii. Google Sheets integration
iv. Metadata extraction

3.3.3 Data Cleaning and Preprocessing Module

This module automatically improves data quality before analysis. It identifies and handles missing values, duplicate records, inconsistent data types, and formatting issues. The cleaned dataset is then prepared for further analytical operations. The main functions of this module are:

i. Missing value treatment

ii. Duplicate removal
iii. Data type correction
iv. Data normalisation

3.3.4 Exploratory Data Analysis (EDA) Module

The EDA Module generates statistical summaries and visual reports that help users understand the characteristics of the dataset. It identifies trends, distributions, correlations, and potential anomalies within the data. The main functions of this module are:

i. Statistical analysis
ii. Correlation analysis
iii. Data visualisation
iv. EDA report generation

3.3.5 AI Query Module

The AI Query Module allows users to interact with datasets using natural language. User questions are processed through the Gemini AI model, which generates analytical responses and visualisations without requiring knowledge of SQL or programming languages. The main functions of this module are:

i. Natural language query processing
ii. AI-generated insights
iii. Interactive analysis
iv. Visualisation generation

3.3.6 AutoML Module

The AutoML Module automates the machine learning process. It detects the appropriate prediction task, trains multiple models using FLAML, evaluates performance, and selects the best-performing model automatically. The main functions of this module are

i. Task detection
ii. Model training
iii. Model evaluation
iv. Feature importance analysis

3.3.7 Insight Generation Module

The Insight Generation Module analyses processed datasets and automatically identifies significant patterns, trends, correlations, and anomalies. The generated insights help users make informed decisions. The main functions of this module are:

i. Trend detection
ii. Correlation analysis
iii. Anomaly identification
iv. Insight ranking

3.3.8 Report Generation Module

This module compiles analytical results into professional reports. It combines visualisations, statistical findings, AI-generated summaries, and recommendations into downloadable HTML or PDF reports. The main functions of this module are:

i. Report creation
ii. Executive summaries
iii. PDF export

3.3.9 Dashboard and Visualisation Module

The Dashboard Module provides a centralised interface for viewing datasets, KPIs, charts, reports, and analytical results. It enables users to monitor project activities and access generated insights efficiently. The main functions of this module are:

i. KPI visualisation
ii. Interactive charts
iii. Activity monitoring
iv. Report access

Together, these modules form an integrated analytics ecosystem capable of handling the complete data analytics lifecycle, from data acquisition and preparation to insight generation, predictive modelling, and report creation. The successful implementation of these modules relies on a combination of modern web technologies, machine learning frameworks, analytical techniques, and cloud services, which are discussed in the following chapter.

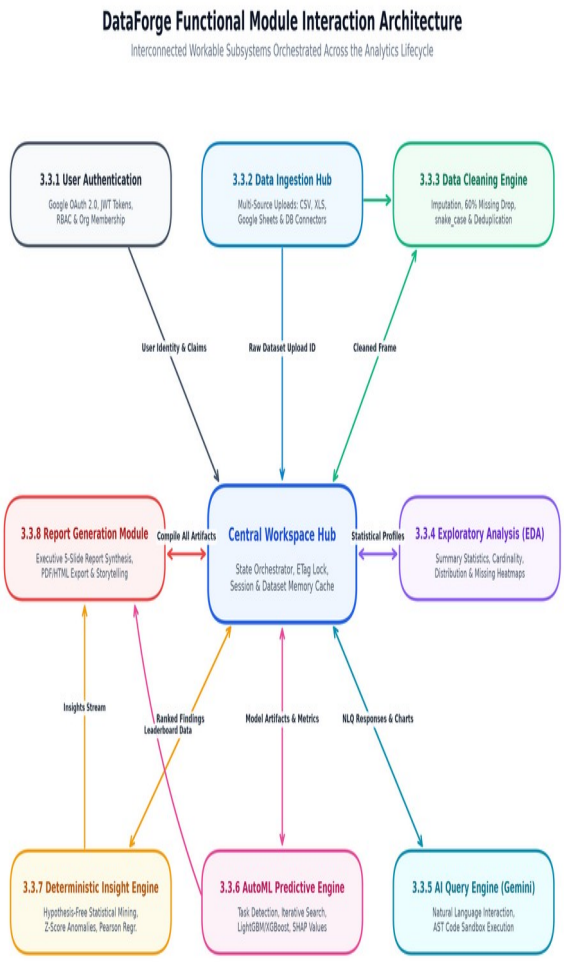


Figure 3.5: DataForge Functional Module Interaction Diagram

Figure 3.5: Functional Module Interaction Diagram

CHAPTER 4

TOOLS AND TECHNIQUES

4.1 Technology Stack

DataForge is developed using a combination of modern web technologies, data analytics frameworks, machine learning libraries, and cloud services. The selected technology stack was chosen to ensure scalability, reliability, ease of development, and efficient processing of analytical workloads. The backend is built using Python-based frameworks, while the frontend provides an interactive and responsive user experience. Advanced analytics and artificial intelligence capabilities are enabled through specialised libraries and cloud-based AI services. Table 4.1 presents the major technologies used in the development of the system along with their respective roles.

Table 4.1: Technology Stack Used

Category	Technology	Purpose
Backend Framework	Flask 3.x (Python)	REST API development, routing, and session management
Database	Supabase PostgreSQL	Storage of users, uploads, reports, jobs and metadata
Object Storage	Supabase Storage	Storage of datasets, models and reports
Authentication	Authlib + Google OAuth 2.0	Secure user authentication
Real-Time Communication	Flask-SocketIO + Socket.IO	Live task updates and notifications
Task Queue	Celery + Redis	Background processing of EDA, AutoML and report generation
AutoML Engine	FLAML	Automated machine learning model training

Data Processing	Pandas, NumPy	Data manipulation and analysis
EDA Engine	ydata-profiling	Automated exploratory data analysis
AI Integration	Google Gemini 2.5 Flash	Natural language query processing and report summarisation
Visualisation	Chart.js	Interactive charts and dashboards
Frontend	HTML5, Jinja2, Bootstrap 5, JavaScript	User interface development

4.2 Algorithms and Techniques

While the technology stack provides the infrastructure required to develop and deploy DataForge, the intelligence of the system is driven by a set of analytical algorithms and techniques. These techniques automate data preparation, insight generation, machine learning, natural language interaction, and report creation, enabling users to transform raw data into meaningful information with minimal manual effort. The major algorithms and methodologies implemented in the system are discussed in the following sections.

4.2.1 Automated Data Cleaning Pipeline

Data quality plays a crucial role in the accuracy and reliability of analytical results. Real-world datasets often contain missing values, duplicate records, inconsistent formatting, and incorrect data types that can negatively affect analysis and model performance. To address these challenges, DataForge incorporates an Automated Data Cleaning Pipeline that automatically detects and resolves common data quality issues before any analytical operation is performed.

The cleaning process begins by examining the uploaded dataset and identifying columns with excessive missing values. Columns containing more than 60% missing data are removed, while the remaining missing values are handled using appropriate statistical imputation techniques. Numerical attributes are filled using mean or median values depending on the distribution, whereas categorical attributes are completed using the mode. The pipeline further removes duplicate records, standardises column names, trims unnecessary whitespace and finally produces a clean dataset suitable for further analysis.

The Data Cleaning Pipeline is responsible for improving data quality before analytical processing. The system automatically detects and resolves common data quality issues such as missing values, duplicate records, inconsistent naming conventions, and invalid data types. The working procedure of the automated data cleaning pipeline is as follows:

1. Load the uploaded dataset.
2. Identify columns with excessive missing values.
3. Remove columns having more than 60% missing data.
4. Impute remaining missing values:
    i. Mean for normally distributed numerical data.
    ii. Median for skewed numerical data.
    iii. Mode for categorical attributes.
5. Remove duplicate records.
6. Standardise column names into snake_case format.
7. Trim unnecessary whitespace.
8. Convert data types to appropriate numeric or datetime formats.
9. Generate a cleaned dataset.

To ensure that the uploaded data is suitable for analytical processing, the Data Cleaning Pipeline follows an automated data cleaning approach. The algorithm systematically identifies and resolves common data quality issues such as missing values, duplicate records, inconsistent column names, and incorrect data types. By applying predefined cleaning rules, the system transforms raw and unstructured data into a standardised and reliable dataset that can be used for further analysis, visualisation, and machine learning tasks. The steps involved in the algorithm are given below:

Input: Raw Dataset `D`

Output: Clean Dataset `C`

1. Load dataset `D`
2. Identify missing values
3. If the missing percentage > 60%
    Remove column
4. Else
    Fill numerical values using mean/median
    Fill categorical values using the mode
5. Remove duplicate rows

6. Standardise column names
7. Convert data types where applicable
8. Return cleaned dataset `C`

4.2.2 FLAML AutoML

Predictive analytics often requires extensive experimentation with different machine learning algorithms and hyperparameter settings. Performing this process manually can be time-consuming and demands significant expertise. To simplify model development, DataForge integrates the FLAML (Fast Lightweight AutoML) framework, which automatically identifies the problem type, selects suitable algorithms, and optimises their performance within a specified time budget.

Once the user selects a target variable, the system determines whether the task is a classification or regression problem. The data is then preprocessed and multiple machine learning models, including LightGBM, XGBoost, Random Forest and Linear Models are trained and evaluated automatically. Based on performance metrics, the most suitable model is selected, interpreted using SHAP values, and stored for future use.

The AutoML engine follows a systematic workflow to automate model selection, training, and evaluation. Starting from the user-selected target variable, the system preprocesses the data, identifies the problem type, evaluates multiple candidate models, and finally selects the best-performing model for predictive analysis. The complete working procedure is outlined below:

1. User selects target variable.
2. System detects task type.
3. Features are encoded and cleaned.
4. Training and testing datasets are generated.
5. FLAML performs iterative model search.
6. The best model is selected.
7. Evaluation metrics are calculated.
8. SHAP values are generated for interpretability.
9. The trained model is stored for future use.

The AutoML engine utilises the FLAML framework to automatically identify the most

suitable machine learning model for the given dataset. The algorithm performs dataset preprocessing, model training, performance evaluation, and model selection within a specified optimisation budget, ensuring an efficient and accurate predictive analytics workflow. The steps involved in the algorithm are given below:

Input: Dataset `D`, Target `T`

Output: Best Model `M`

1. Detect task type
2. Preprocess dataset
3. Split into train and test sets
4. Initialise FLAML
5. For available time budget:
    Train candidate models
    Evaluate performance
    Update the best model
6. Select the highest-performing model
7. Compute evaluation metrics
8. Generate feature importance
9. Save model `M`

4.2.3 Deterministic Insight Engine

The **Deterministic Insight Engine** is designed to automatically extract meaningful and actionable insights from datasets without requiring extensive user intervention. Since large datasets often contain hidden patterns that are difficult for non-technical users to identify, the engine applies a combination of statistical techniques and rule-based analytical methods to uncover trends, anomalies, correlations, and distribution characteristics. These insights help users understand the underlying behaviour of their data and support decision-making.

To achieve this, the engine utilises several analytical detectors. Trend Detection employs linear regression techniques to identify increasing or decreasing trends within numerical variables over time. This allows the system to recognise whether a particular metric is growing, declining, or remaining stable.

The engine also performs Anomaly Detection, which is used to identify unusual observations that significantly deviate from normal data behaviour. For this purpose, the system applies the Z-score method, which measures how far a data point lies from the mean in terms of standard

deviations. The Z-score is calculated as:

where:

- = observed value
- = mean of the dataset
- = standard deviation

Values having are considered anomalies and are flagged as potential outliers.

Another important analytical component is Correlation Analysis, which determines the strength and direction of the relationship between two numerical variables. The system uses the Pearson Correlation Coefficient, calculated as:

where:

- = individual observations
- = mean values of the variables
- = correlation coefficient

A strong relationship is identified when , indicating that the variables exhibit a significant positive or negative association.

To further understand the characteristics of the dataset, the engine performs Distribution Analysis. This process evaluates statistical measures such as Mean, Median, Standard Deviation, Skewness, Kurtosis, and Interquartile Range (IQR). These metrics provide information about the central tendency, variability, symmetry, spread, and overall shape of the data distribution. By analysing these measures, the system can identify whether the data follows a normal distribution, contains significant variability, or exhibits unusual behavioural patterns.

The overall insight generation process is carried out through an automated algorithm that sequentially applies these analytical techniques. The system first examines the dataset structure, executes trend, anomaly, correlation, contribution, and distribution analyses, and then evaluates the generated observations. Finally, the discovered insights are ranked according to their significance and presented to the user in an easily understandable form. The

steps involved in the insight generation process are shown below:

Input: Dataset D

Output: Ranked Insights I

1. *Detect dataset schema*
2. *Execute trend analysis*
3. *Execute anomaly detection*
4. *Execute correlation analysis*
5. *Execute contribution analysis*
6. *Execute distribution analysis*
7. *Rank insights by importance*
8. *Return top insights*

After the dataset has been cleaned and analysed, DataForge automatically extracts meaningful patterns and observations through the Deterministic Insight Engine. This component applies statistical and rule-based analytical techniques to uncover trends, anomalies, correlations, and distribution characteristics that support data-driven decision-making.

4.2.4 Natural-Language Query Pipeline

While statistical analysis provides valuable insights, many users may not possess the technical expertise required to explore datasets through conventional analytical methods. To make data exploration more accessible, DataForge provides a Natural Language Query (NLQ) interface that enables users to interact with datasets using plain English queries.

The module integrates Google Gemini 2.5 Flash to interpret user queries, analyse dataset structure, generate appropriate analytical operations, and return responses in the form of charts, tables, and natural language summaries.

The complete working procedure of Natural language query pipelines is as follows :

1. User enters a natural language query.
2. Dataset schema and sample rows are extracted.
3. Prompt is generated.
4. Prompt is sent to Gemini 2.5 Flash.
5. Gemini returns:

i. Analysis logic
ii. Visualisation intent
iii. Natural language response

6. Generated code is validated through an AST security layer.
7. Safe code is executed on the dataset.
8. Results are returned as charts, tables, or summaries.

To enable conversational interaction with datasets, the Natural Language Query Processing algorithm interprets user queries, analyses their intent, retrieves relevant dataset information, and generates meaningful responses using the integrated AI model. The algorithm follows a sequence of processing steps, beginning with query interpretation and context extraction, followed by information retrieval and response generation. The steps involved in the algorithm are presented below:

Input: Query `Q`, Dataset `D`
Output: Response `R`

1. Read user query `Q`
2. Extract schema information
3. Generate prompt `P`
4. Send `P` to Gemini API
5. Receive generated analysis
6. Validate generated code
7. Execute approved code
8. Generate visualisation
9. Return response `R`

4.2.5 Report Generation Technique

The final stage of the analytics workflow focuses on communicating generated insights in a clear, structured, and decision-ready format. To support effective interpretation and presentation, DataForge incorporates an automated Report Generation module that compiles analytical findings, visualisations, and machine learning outputs into professional business reports.

This module integrates multiple sources of analytical output, including dataset statistics,

results from the Insight Engine, AutoML evaluation metrics, and AI-generated executive summaries. By combining these components, it produces a structured five-slide analytical report designed for business users and decision-makers.

Specifically, the module brings together:

i. Dataset statistics

ii. Insight Engine outputs

iii. AutoML results

iv. Gemini-generated executive summaries

to generate a coherent and standardised report structure.

The generated report organises insights into a consistent format that enhances interpretability and supports data-driven decision-making. It typically includes an overview of the dataset, key analytical findings, data quality assessment, and actionable recommendations. The final report is structured into the following sections:

1. Cover Page
2. Executive Summary
3. Key Insights
4. Data Quality Assessment
5. Recommendations

The report generation process is implemented through an automated pipeline that systematically collects analytical outputs, processes them into meaningful summaries, and assembles them into a predefined structure. The steps involved in this process are as follows:

Input: Insights, Statistics, AutoML Results

Output: Structured Business Report

1. *Collect analytical outputs from all modules*
2. *Generate KPI and statistical summaries*
3. *Create relevant visualisations for interpretation*
4. *Generate AI-based executive commentary*
5. *Assemble all components into predefined report sections*
6. *Export the final report in presentation format*

This structured procedure ensures consistency, reduces manual effort, and enables rapid transformation of raw analytical results into actionable business intelligence.

After the architecture, algorithms, and analytical techniques used in DataForge, it is essential to identify the hardware and software resources required for the effective implementation and

operation of the system.

4.3 Hardware and Software Requirements

The successful operation of DataForge requires a combination of suitable hardware resources and software technologies. Since DataForge is a web-based data analytics platform, it does not require specialised hardware and can operate on standard computing devices. The system leverages modern software frameworks, machine learning libraries, database services, and cloud technologies to provide data ingestion, preprocessing, exploratory data analysis, automated machine learning, natural language querying, and report generation capabilities.

The hardware requirements define the minimum computing resources necessary for executing the application efficiently, while the software requirements specify the technologies, frameworks, libraries, and development tools used during the implementation and deployment of the system.

4.3.1 Hardware Requirements

The system does not require any specialised hardware components. The minimum hardware requirements are as follows:

i. A computer or laptop with standard processing capability

ii. Minimum 4 GB RAM (8 GB or higher recommended for datasets with large volumes of data)

iii. At least 2 GB of available disk space for application files, datasets, generated reports, and trained machine learning models

iv. Stable internet connectivity for accessing Google Sheets, Google OAuth authentication, Gemini AI services, and cloud storage resources

v. Standard input/output devices such as keyboard, mouse, and monitor

vi. Since DataForge is implemented as a browser-accessible web application, it can be accessed from any device capable of running a modern web browser, including desktop computers and laptops.

4.3.2 Software Requirements

DataForge is developed using Python and integrates multiple open-source libraries, frameworks, and cloud-based services to provide data analytics, machine learning, artificial intelligence, real-time communication, report generation, and user authentication functionalities. The development environment combines backend technologies, machine learning frameworks, visualisation libraries, database services, and task scheduling tools to ensure efficient processing and scalability of the platform. These software components collectively support data ingestion, automated preprocessing, exploratory data analysis, natural language interaction, predictive modelling, dashboard visualisation, and report generation. Furthermore, cloud services and asynchronous processing mechanisms are utilised to enhance system responsiveness and enable efficient handling of computationally intensive analytical tasks. The software requirements and development environment used for the implementation of the proposed system are presented in Table 4.2.

Table 4.2: Software Requirements and Development Environment

Software / Library	Version / Notes
Python	3.10 or higher
Flask	3.x
Flask-Login	0.6.x
Flask-SocketIO	5.x
Authlib	1.x (Google OAuth 2.0)
Pandas	2.x
NumPy	1.x / 2.x
FLAML	2.x
ydata-profiling	4.x
scikit-learn	1.x
LightGBM / XGBoost	Latest Stable Release
Google Gemini API SDK	Latest Stable Release
Supabase Python SDK	2.x

Redis	Latest Stable Release
Celery	5.x
Celery Beat	5.x
Chart.js	4.x
Bootstrap 5	Latest Stable Release
JavaScript (ES6+)	Frontend Functionality
HTML5 / Jinja2	User Interface Templates
Google Chrome / Mozilla Firefox / Microsoft Edge	Version 110+

CHAPTER 5
SYSTEM IMPLEMENTATION AND RESULTS

This chapter presents the implementation and experimental results of the DataForge platform. The system was developed as a web-based analytics solution that integrates dataset management, automated data cleaning, exploratory data analysis, AI-powered querying, insight generation, and automated machine learning within a unified environment.

The implementation focuses on providing a user-friendly interface that enables users to perform complex analytical tasks without requiring extensive technical knowledge. Various modules were developed and integrated to ensure seamless data processing, visualisation, predictive modelling, and report generation. The following sections describe the implementation of each major component of the system and demonstrate the results obtained through its operation.

5.1 User Authentication

The User Authentication module provides secure access to the platform using Google OAuth 2.0. It enables users to log in safely and access their personalised analytics workspace. This ensures secure management of datasets and generated reports.

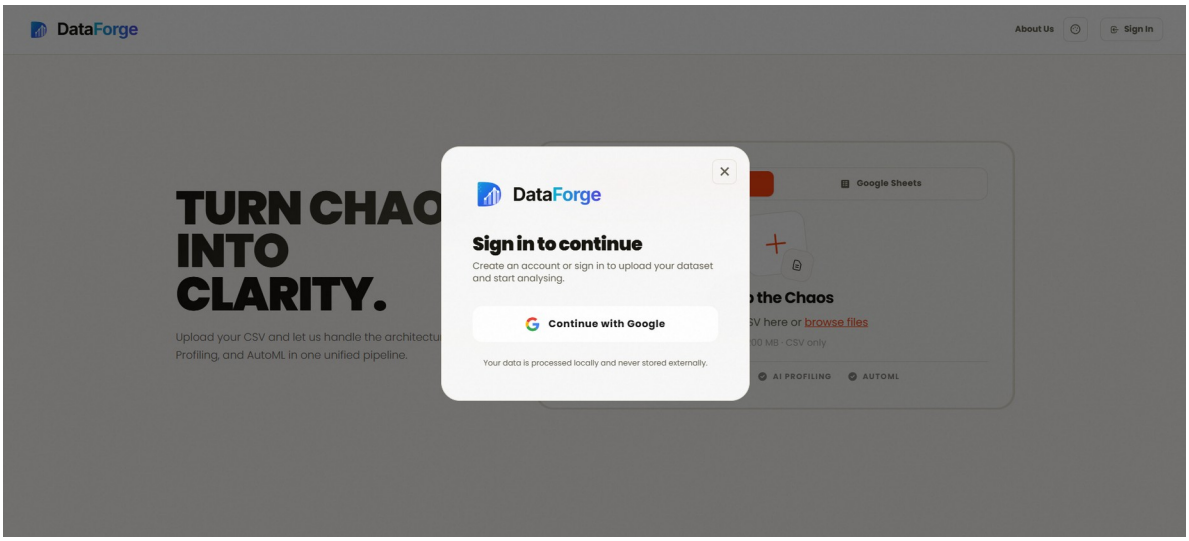


Figure 5.1: User Login Interface

5.2 Dataset Upload Module

The Dataset Upload module allows users to import data from CSV, XLS, XLSX files, or Google Sheets. The system validates the uploaded dataset and prepares it for further analysis. This provides a flexible and user-friendly data ingestion process.

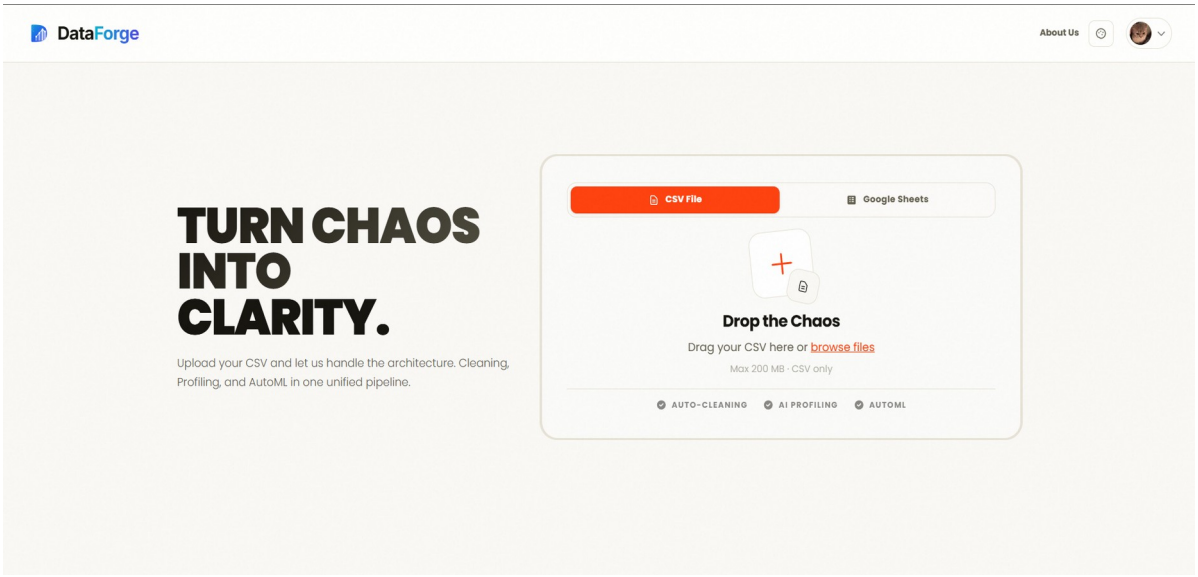


Figure 5.2: Dataset Upload and Import Interface

5.3 Data Cleaning Module

The Data Cleaning module automatically detects and resolves common data quality issues such as missing values, duplicate records, and inconsistent data types. The cleaned dataset improves the reliability and accuracy of subsequent analytical operations.

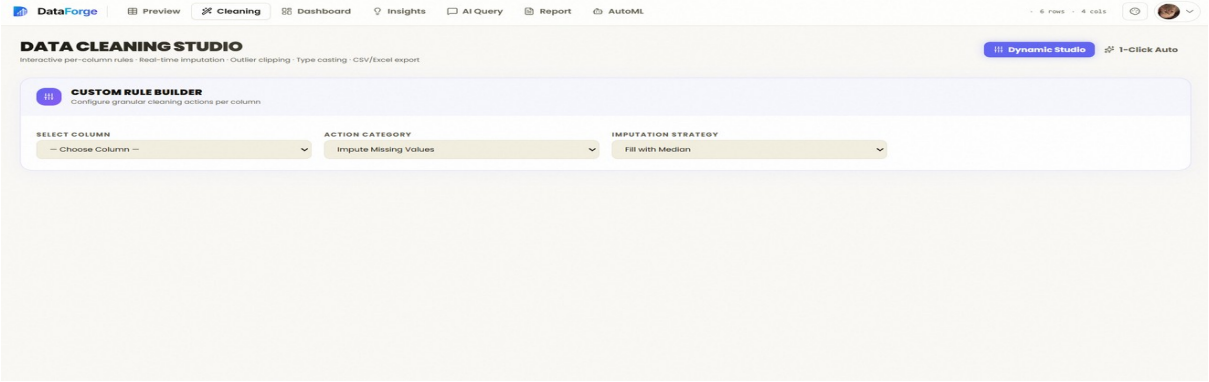


Figure 5.3: Data Cleaning Results

5.4 Exploratory Data Analysis (EDA)

The EDA module generates statistical summaries and visualisations to help users understand their datasets. It provides information about distributions, correlations, and key characteristics of the data. This enables better understanding before advanced analysis is performed.

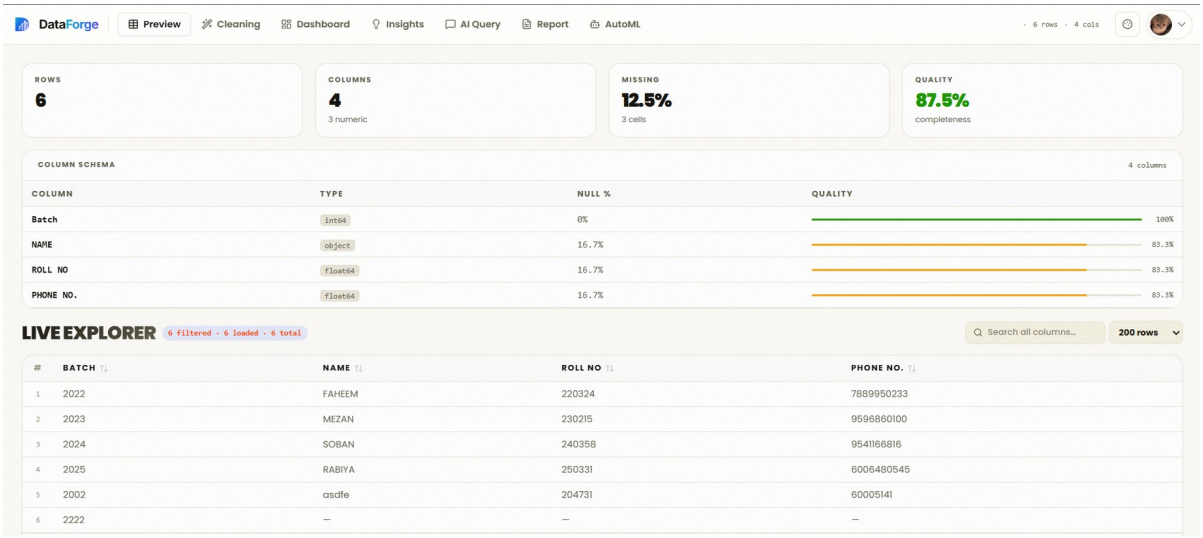


Figure 5.4: EDA Report

5.5 AI Query Module

The AI Query module allows users to interact with datasets using natural language questions. The system interprets user queries and generates meaningful responses along with supporting visualisations. This simplifies data exploration for non-technical users.

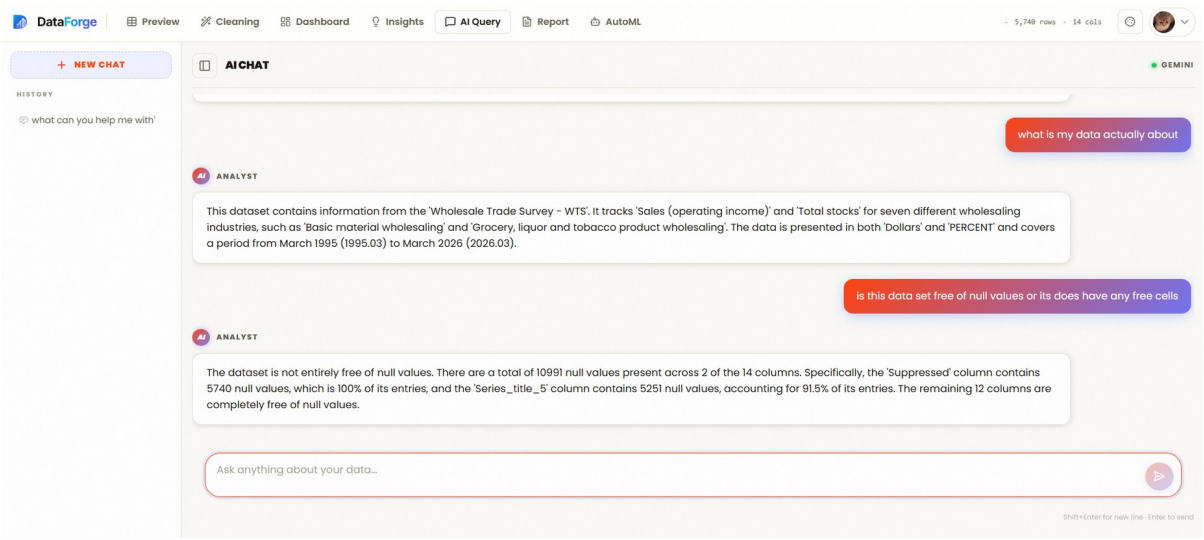


Figure 5.5: AI Query Interface

5.6 AutoML Module

The AutoML module automatically trains and evaluates multiple machine learning models. It selects the best-performing model based on evaluation metrics and presents the results to the user. This eliminates the need for manual model selection and tuning.

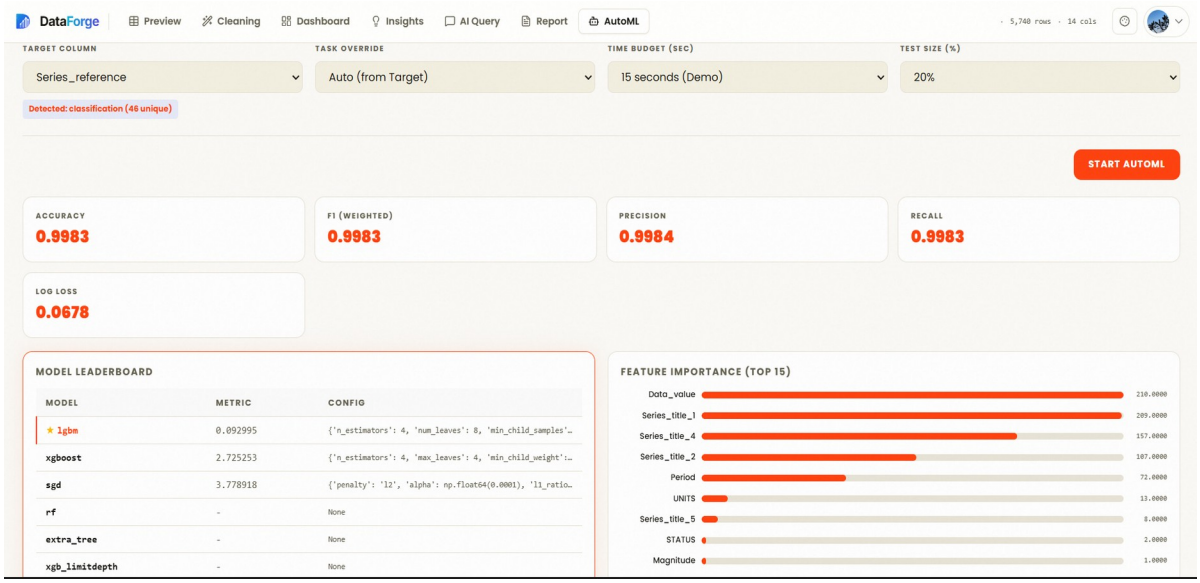


Figure 5.6: AutoML Results

5.7 Insight Generation Module

The Insight Generation module automatically identifies trends, correlations, anomalies, and important patterns within the dataset. The generated insights help users understand key findings and support data-driven decision-making.

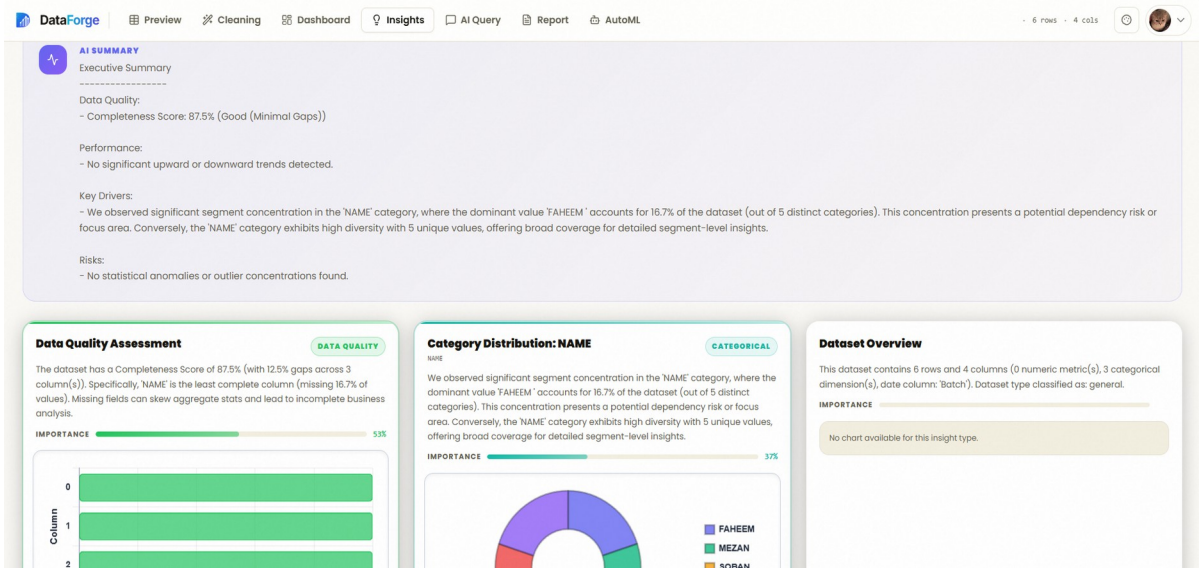


Figure 5.7: Generated Insights

5.8 Dashboard and Monitoring

The Dashboard provides a consolidated view of analytical activities, key metrics, and

generated reports. It enables users to monitor analysis results and track project progress through an interactive and user-friendly interface.

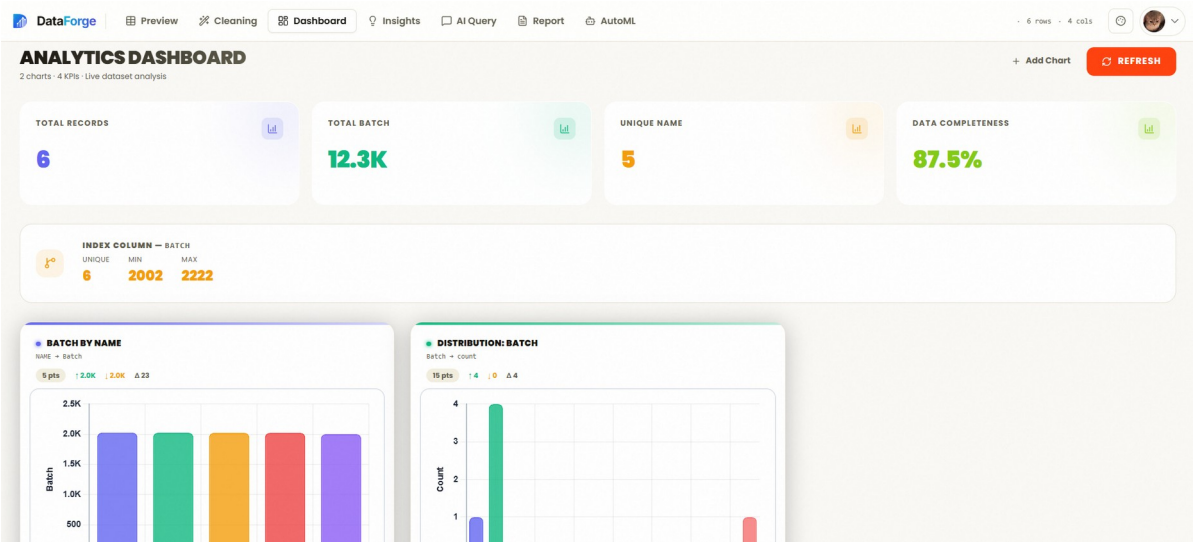


Figure 5.8: Dashboard Interface

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

DataForge presents an integrated approach to democratising data analytics for non- technical users by consolidating the entire data science lifecycle data ingestion, automated cleaning, exploratory analysis, AutoML-based model training, natural- language AI querying, insight detection, and automated reporting within a single web application. By abstracting the complexity of programming environments and fragmented analytics tools, the platform enables analysts, researchers, and students to extract meaningful insights from structured datasets with minimal technical overhead.

The system architecture demonstrates that a lightweight Flask-based framework, combined with modern open-source libraries such as FLAML, ydata-profiling, and Flask-SocketIO, and supported by cloud services like Google Gemini and Supabase, can deliver capabilities comparable to enterprise analytics platforms at significantly lower infrastructure cost. The deterministic insight engine further ensures system reliability by providing meaningful analytical outputs even when external AI services are unavailable.

6.2 Future Work

Most of the gaps that have been identified during the literature review have been addressed through the development of the DataForge platform. The system successfully integrates automated data cleaning, exploratory data analysis, AI-powered querying, insight generation, automated machine learning, and report generation within a unified environment. However, there are still several areas where the platform can be further enhanced to improve scalability, flexibility, and analytical capabilities.

1. **Migration to Next.js and FastAPI**
   Upgrade the technology stack to improve performance, scalability, and API response times. This will enhance the overall user experience and support larger workloads.
2. **Advanced Forecasting and Predictive Analytics**
   Implement advanced forecasting techniques and time-series models for trend

prediction and future value estimation. This will provide deeper analytical capabilities for decision-making.

3. **Multi-Sheet Spreadsheet and Multi-Table Synthesis**

   Enable automatic analysis and integration of multiple spreadsheet sheets and related database tables. This will help users generate insights from interconnected datasets.

4. **Semi-Structured Data Analytics**

   Extend support to JSON and XML. This will allow the platform to handle a wider variety of real-world data formats.

5. **Dynamic Data Cleaning**

   Incorporate intelligent data cleaning mechanisms that automatically adapt to different dataset characteristics. This will improve data quality while reducing manual preprocessing efforts.

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